EXPERIMENT 4: POLYALPHABETIC SUBSTITUTION CIPHER

1. Aim

To write and execute a C program to implement the polyalphabetic substitution cipher using a repeating keyword.

2. Algorithm

1. Start the program.
2. Initialize the plaintext and keyword.
3. Repeat the keyword to match the plaintext length.
4. Shift each plaintext letter according to the corresponding keyword letter.
5. Display the encrypted text.
6. Stop the program.

3. CODE

```
import subprocess

code = r'''
#include <stdio.h>

int main()
{
    char text[] = "HELLO";
    char key[] = "KEY";
    int i, shift;

    printf("Plaintext: %s\n", text);
    printf("Key: %s\n", key);

    for (i = 0; text[i] != '\0'; i++)
    {
        shift = key[i % 3] - 'A';
        text[i] = (text[i] - 'A' + shift) % 26 + 'A';
    }

    printf("Ciphertext: %s\n", text);

    return 0;
}
'''

with open("polyalphabetic.c", "w") as f:
```

```
    printf("Ciphertext: %s\n", text);

    return 0;
}
'''

with open("polyalphabetic.c", "w") as f:
    f.write(code)

compile_result = subprocess.run(
    ["gcc", "polyalphabetic.c", "-o", "polyalphabetic"],
    capture_output=True, text=True
)

if compile_result.returncode != 0:
    print(compile_result.stderr)
else:
    result = subprocess.run(
        ["./polyalphabetic"],
        capture_output=True, text=True
    )

    print(result.stdout)

    with open("polyalphabetic_output.txt", "w") as f:
        f.write(result.stdout)
```

OUTPUT:

```
Plaintext: HELLO
Key: KEY
Ciphertext: RIJVS
```

RESULT:

Thus, the C program for the polyalphabetic substitution cipher was implemented and executed successfully, encrypting the plaintext HELLO into the ciphertext RIJVS using the keyword KEY.